Samanga Dwivedi

ma24btech11021@iith.ac.in | samanga2006@gmail.com | +91 9510823366 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.51
RMG Maheshwari English School	Class XII (CBSE)	2024	92.20%
Delhi Public School Surat	Class X (CBSE)	2022	97.60%

RELEVANT COURSEWORK

Courses

- **Computer Science:** DSA, Algorithms, Theory of Computation, Introduction to Programming, DBMS, Compilers
- **Mathematics:** Number Theory, Linear Algebra, Probability Theory, Applied Statistics, Convex Optimization, Ordinary Differential Equations, Linear Systems and Signal Processing

Self-taught

- Advanced Learning Algorithms, Supervised Machine Learning
- Unsupervised Learning, Recommenders, Reinforcement Learning
- Manifold Learning and Dimensionality Reduction

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, SQL, JavaScript, Latex
- **Tools & Technologies:** Linux, Git, GitHub, VS Code, LaTeX
- **Libraries/Frameworks:** NumPy, SciPy, Pandas, TensorFlow, Matplotlib

PROJECTS

Private BitTorrent Platform | Github

- Engineered a **BitTorrent-based P2P file-sharing system**, implementing bencoding, info_hash computation, tracker communication, peer handshakes, block-level piece downloading, **SHA-1 integrity verification, and asyncio-based concurrent multi-peer transfers**.
- Built a **FastAPI-based custom tracker and torrent indexing service** with automated .torrent generation, SQLite-backed persistent peer management, compact peer-list encoding, dynamic swarm discovery, and automated stale-peer cleanup.

Real-Time Auction Website | Github

- Engineered a **real-time auction platform** using React, Node.js, Express, and Socket.IO, enabling synchronized live bidding and auction-state updates across concurrent participants.
- Implemented **bid validation, automated winner determination, and dynamic auction control**, with PostgreSQL and Prisma ORM for persistent team, question, and auction-state management.
- Built **JSON Web Tokens-based admin authentication** and a responsive dashboard for managing teams, questions, live auction states, and analytics.

Voronoi Diagram (Dr. Balasubramaniam Jayaram) | Github

- Implemented **Fortune's sweep-line algorithm** for Voronoi diagram construction, using a **priority-queue event scheduler** and **binary-tree beach line** to process site and circle events.
- Implemented dynamic **arc insertion/removal, edge-intersection detection, and Voronoi cell construction**, enabling nearest-neighbor point location with Euclidean-distance verification.

Statistical Modelling and Hypothesis Testing | Gdrive

- Applied **statistical estimation and inference techniques**, including sampling distributions, point/interval estimation,

MoM, MLE, and confidence intervals across real-world datasets.
- Performed **one-/two-tailed t-tests and ² tests** to evaluate statistical hypotheses, and built **linear regression models** using least-squares estimation, significance testing, and R^2-based evaluation.

EXPERIENCE AND PORS

Undergraduate Researcher (with Dr. Balasubramaniam Jayaram) | Github

- Investigating the impact of outliers on **distance-based and manifold-learning embeddings**, analyzing structural distortion and **neighborhood preservation** under dimensionality reduction.
- Evaluated **MDS, t-SNE, and UMAP** across biological datasets, systematically studying the effects of hyperparameters and outlier positioning on low-dimensional representations.
- Developed Python-based experimental pipelines using **NumPy, Pandas, Scikit-learn, UMAP, and Matplotlib** for quantitative evaluation and visualization.

Teaching Assistant - **Series of Functions**, Dr. Balasubrahmaniam Jayaram

- Designed rigorous assignments and problem sets on uniform convergence, power series, and Taylor/Fourier series under faculty guidance.
- Administered course examinations, evaluating undergraduate proof-writing skills.

Software Development Core, Tinkerer's Lab

- Developed a real-time auction platform 2 for Tinkerers' Lab Thrust event, enabling competitive bidding with dynamic price updates and user interaction, handling concurrent bids and ensuring consistency, and efficient state management during live auctions
- Conducted hands-on sessions on HTML, CSS, and React during Tinker Camp, mentoring participants in building web applications

Assistant Manager, E-cell

- Led organization of flagship events E-Summit and E-Merge at Entrepreneurship Cell, overseeing logistics, team coordination, and participant engagement